

ECE 102
B. Tech. IInd SEMESTER EXAMINATION, 2024-25
BACHELOR OF TECHNOLOGY
(IT & ECE)
Basic Electrical Engineering

Time : Three Hours]

[Maximum Marks : 75

Note: There are **three** sections (A, B and C) and Candidate has to attempt question from all sections. Marks are indicated against each section.

Section A

1. Attempt all Questions. 5×3=15
- (a) Explain the following :
- (i) Unilateral and bilateral elements.
 - (ii) Linear and Non-linear elements.
- (b) Compute equivalent resistance of three resistance connected in
- (i) Series
 - (ii) Parallel
- (c) Draw and describe power triangle, explaining active, reactive and apparent power.
- (d) State maximum power transfer theorem.
- (e) Define form factor and peak factor.

Section-B

Note : Attempt All Questions.

$4 \times 5 = 20$

2. (a) Explain voltage and current source of a network. Explain source transformation principle in any circuit.

Or

- (b) Calculate current in all resistances of the circuit shown in figure (1) using nodal analysis.

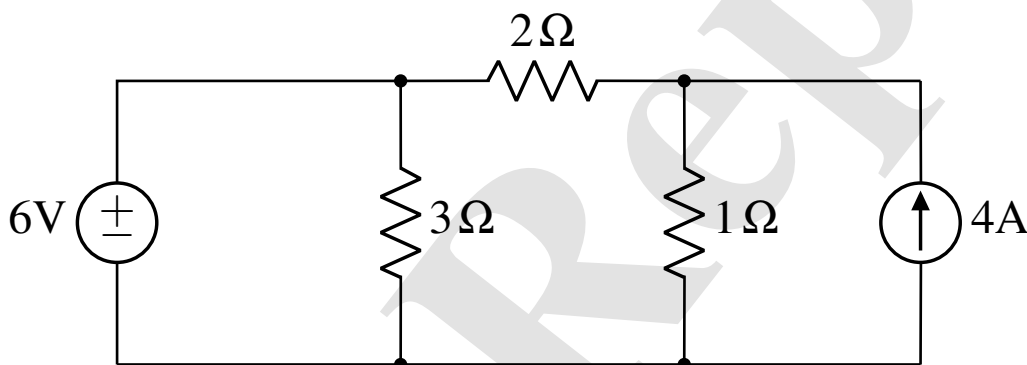


Fig. (1)

3. (a) Find the Thevenin's equivalent circuit of the circuit shown in figure (2) and also find the current through $R_L = 6\Omega$.

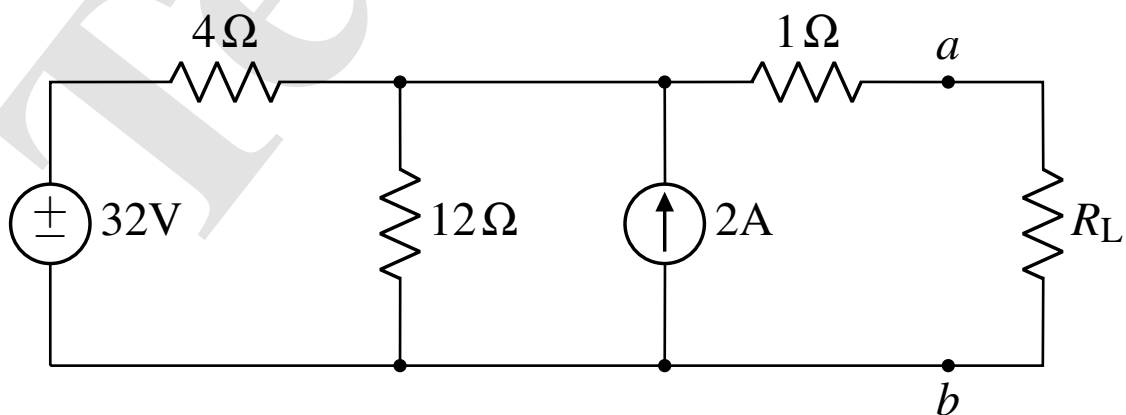


Fig. (2)

Or

- (b) Define resonance and derive resonance frequency of parallel RLC circuit also give the graphical representation of the circuit.

4. (a) Explain two wattmeter method for power measurement in three phase system.

Or

- (b) A PMMC instrument gives a reading of 25 mA when the potential difference across its terminals is 75 mV, calculate shunt resistance for full scale deflection corresponding to 50 A.

5. (a) Explain B-H curve of a magnetic material.

Or

- (b) Calculate the voltage induced in the armature winding of a 4 pole, wave wound, dc machine having 728 conductors and running at 1800 rpm. The flux per pole is 35 mwb.

Section-C

Note : Attempt any two questions.

2×20=40

6. Derive the expression for average and rms value of sinusoidal waveform.

7. State and explain superposition theorem and find the current 'I' in the following circuit shown in figure (3).

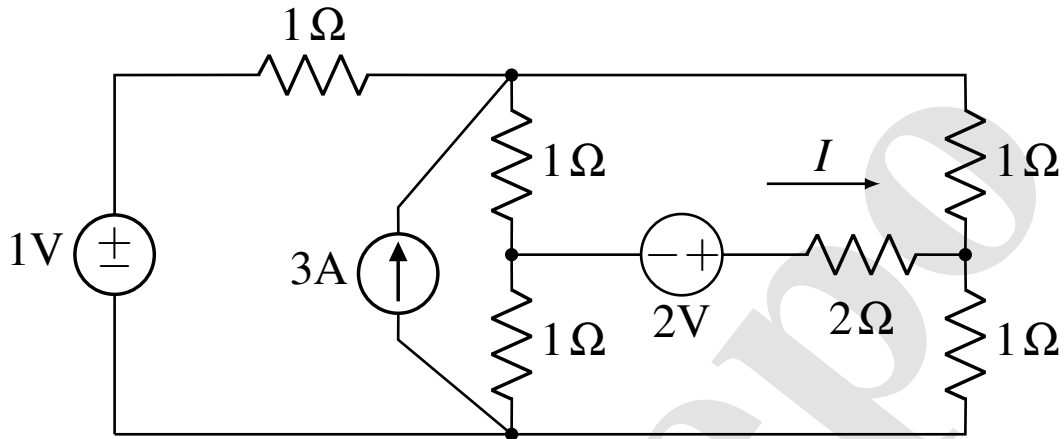


Fig. (3)

8. Describe the construction and working principle of PMMC with the help of diagram.
9. Describe the construction and working principle of transformer also derive the emf equation.
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